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EE1060: Differential Equations and Transform Techniques

Lecture 16: Second-order nonhomogeneous ODEs

Topics :

1. Method of Undetermined Coefficients
2. Introduction to Method of Variation of Parameters

Quick Recall - Annihilator and second-order constant coefficient ODE

$g(x)$	Annihilator
$\sum_{i=1}^{n} a_i x^i$	D^{n+1}
$\sum_{i=1}^{n} a_i x^i e^{ax}$	$(D-a)^{n+1}$
$\sum_{i=1}^{n} a_i x^i e^{ax} \cos mx$	$((D-a)^2+m^2)^{n+1}$

Change in Notation :-

$L[y] = g(x)$

$\underbrace{P(D)}_{\text{function of } D.} y = g(x)$

Annihilator for $g(x) = \phi(D)$

$\Rightarrow \phi(D)\, g(x) = 0.$

Solution of homogeneous Equation: $y'' + Py' + Qy = 0$

$P(D) = (D^2 + PD + Q) \longrightarrow$ roots $= s_1$ & s_2.

roots are real & distinct $\longrightarrow$ $y_c = c_1 e^{s_1 x} + c_2 e^{s_2 x}$

" & repeating $\longrightarrow$ $y_c = (c_1 + c_2 x) e^{s_1 x}$ $\quad [c_1 y_1 + c_2 y_2]$ $\quad y_1 = e^{s_1 x}$, $y_2 = x e^{s_1 x}$

Complex Conjugate $\longrightarrow$ $y_c = Re^{-\sigma x} \cos(\omega_d x + \phi)$ $\quad s = -\sigma + j\omega_d$

$= e^{-\sigma x}(c_1 \cos \omega_d x + c_2 \sin \omega_d x)$

Method of undetermined coefficients - Basic idea

Known : given $L[y] = g(x)$ $\quad P(D)\,y = g(x) \longrightarrow (1)$

Step 1: Compute Annihilator $\varphi(D)$ for $g(x)$

Step 2:- Compute the Complementary sol $L[y]=0$. (y_c)

$$\overbrace{\varphi(D)\,\underbrace{P(D)}\,y} = \varphi(D)\,g(x)$$ $\longrightarrow$ to change the order of operation $P(D)$ & $\varphi(D)$ are constant coefficients

$R(D)\,y = 0 \longrightarrow$ higher order homogeneous ODE

$P(D) \longrightarrow$ Polynomial in D of order n

$\varphi(D) \longrightarrow$ " " " of " m

then $R(D) \longrightarrow m+n$.

Generalization :- $L[y]=0$ order $\rightarrow n$.

roots of auxillary Eqn.

a) real & distinct :- $y(x) = c_1 e^{s_1 x} + \dots + c_n e^{s_n x}$.

b) real & repeat $(s_1, \dots, s_{i-1}, \underbrace{s_i, \dots s_i}_{m \text{ times}})$: $y(x) = c_1 e^{s_1 x} + \dots + c_{i-1} e^{s_{i-1} x} + (c_i + c_{i+1} x + c_{i+2} x^2 + \dots c_{i+m-1} x^{m-1})\, e^{s_i x}$

Method of undetermined coefficients

c) Complex & repeat :- $S_i \rightarrow$ 'm' times.

$$\underbrace{\qquad\qquad}_{\text{other roots}} + e^{-\sigma x}\left[(c_i + c_{i+1}x + c_{i+2}x^2 + \dots + c_{i+m-1}x^{m-1}) \cos mx + (\qquad) \sin mx \right]$$

$R(D) = 0. \Rightarrow Q(D)\,P(D)\,y = 0 \Rightarrow y = \underbrace{\sum_{i=1}^{m+n} c_i \underbrace{y_i(x)}}$ $\longrightarrow$ fundamental solution.

$$= \underbrace{\sum_{i=1}^{n} c_i y_i(x)}_{y_c} + \sum_{i=n+1}^{n+m} c_i y_i(x)$$

$Q(D) \rightarrow (D^2 + m^2) \longrightarrow g(x) = \sin mx.$

$P(D) = (D^2 + K^2) \longrightarrow y'' + k^2 y = \sin mx.$

$R(D) = (D^2 + k^2)(D^2 + m^2) \longrightarrow \pm jm, \pm jk.$

$$y = y_c + \sum_{i=n+1}^{n+m} c_i y_i(x) \longrightarrow (1)$$

$$P(D)\,y = 0 + P(D)\left[\sum_{i=n+1}^{n+m} c_i y_i(x)\right] = g(x)$$

Pick c_i such that $P(D) \underbrace{\sum_{i=n+1}^{n+m} c_i y_i(x)} = g(x)$

$\Downarrow$

y_p.

$y = y_c + y_p \longrightarrow$

Example

find the general solution of $\underbrace{y'' + k^2 y}$ = Sin bx (assume $k \neq b$)
(from Simmon's book)

$P(D) = D^2 + k^2$, $g(x) = \sin bx$

$\varphi(D) = D^2 + b^2$

$R(D) = \underbrace{(D^2+b^2)}(D^2+k^2)$ Try a lin $R_1 \cos(kx+\phi_1) + R_2 \cos(bx+\phi_2)$

when $k=b$,

$(R_1 + R_2 x)\cos(kx+\phi)$

roots $= \pm jb, \pm jk.$ $y = \underbrace{(c_1 \cos kx + c_2 \sin kx)}_{y_c} + \underbrace{(c_3 \cos bx + c_4 \sin bx)}_{\text{I}}$

Pick c_3 & c_4 such that I satisfies $P(D)\,\text{I} = g(x)$.

$(-b^2 + k^2)c_3 \cos bx + (k^2 - b^2)c_4 \sin bx = \sin bx$

$c_3 = 0$

$c_4 = 1/(k^2 - b^2)$

$y(x)$ for $L[y] = g(x) \Rightarrow c_1 \cos kx + c_2 \sin kx + \frac{1}{k^2 - b^2} \sin bx.$

Method of Variation of Parameters

given $L[y] = g(x)$ goal : $y = y_c + y_p$

Requirement : ① y_c is known

→ for Constant Coefficient ODE, y_c can be computed

→ otherwise y_c must be predicted or given.

Solution :- try to find $y_p = \sum v_i(x)\, y_i(x)$ where $y_i(x)$ is a fundamental solution to $L[y]=0$.